DT-GSK in Plain Language

What it is, how it works, and what it proves

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A seven-page companion to the paper *DT-GSK: Dimension-Tiered Adaptive Control and Deterministic Refinement for Gaining-Sharing Knowledge Optimization.*

For readers who already know population-based metaheuristics. Where this summary could be read as claiming more than the paper does, the paper governs.

1. Where this sits: GSK and its descendants

Gaining-Sharing Knowledge (GSK) is a population-based metaheuristic whose update is built on a two-phase human-learning metaphor. Each individual splits its coordinates into a *junior* part, which draws knowledge from nearest and farthest neighbours in the population, and a *senior* part, which draws from the best, middle and worst performers. The junior/senior split is not fixed: a decaying schedule governed by K_{exp} moves coordinates from junior to senior as the run proceeds, so the search shifts from local imitation to elite-guided refinement. Two scalars control the transfer — the knowledge factor KF (step magnitude) and the knowledge ratio KR (how many coordinates actually change) — and selection is greedy.

What the descendants changed

The published variants attack the base method at three levels, and it is worth being precise about which, because it defines the gap.

- **AGSK** added the family's first adaptive parameter control: each individual draws its (KF, KR) pair from a four-setting pool whose probabilities update from realized improvements, plus a dual-regime knowledge rate and L-SHADE-style linear population reduction.
- **APGSK** extended that with a second, negative-KF pool activated after half the budget to re-diversify, a nonlinear reduction schedule, and a much larger initial population.
- **FDB-AGSK** left the parameters alone and redesigned *donor policy*: the senior-phase guide becomes fitness–distance-balance selection, aimed at AGSK's premature convergence on multimodal, hybrid and composition functions.
- **ATMALS-GSK** and **eGSK** are composite designs, adding memory-based local search and a dual adaptive knowledge factor with a gradient-based interior-point refinement respectively.

Two things are common to all of them. What adapts is a *scalar* — a parameter value, or which individual knowledge flows from — never *which coordinates move together*. And each variant fixes its behaviour at a single operating point.

The gap

Every published variant uses one configuration regardless of problem dimension. That is a real weakness, because the useful machinery changes qualitatively with D.

At $D = 10$ the space is small enough that careful structure learning and a deterministic endgame are affordable and pay off. At $D = 1000$ the same machinery consumes budget that a plain, well-scheduled population would spend better — while controllers that are pointless at low D (basin memory, subspace floors, acceptance clipping) start to matter. Evidence bases compound this: AGSK and APGSK are evaluated at $D \leq 20$ under budgets far larger relative to dimension than the CEC2017 protocol, and FDB-AGSK at a tenth of the standard budget with $D = 10$ untested.

One operating point cannot be right across that range.

2. What DT-GSK does differently

DT-GSK resolves control *by dimension* rather than at one operating point. It defines four tiers — $D < 20$, 20–49, 50–99, $D \geq 100$ — and gates whole subsystems on or off across them.

The tier is resolved once, from D, before the run starts, and never changes during it. There is no online tier switching to reason about: a $D = 60$ problem simply instantiates a different machine than a $D = 15$ one.

Subsystem	$D<20$	20–49	50–99	$D\geq100$
Inherited GSK core (mask and ties modified)	on	on	on	on
NLPSR population-size reduction	on	on	on	on
ACE operator selector + ARGP pruning	on	on	on	on
Linkage block crossover	on	on	on	on
Coordinate local search	on	on	on	on
BSE escape + archive, deep-stall restart	on	on	on	on
ISM interaction memory	off	off	**on**	**on**
Eigenframe final polish	off	off	**on**	**on**
Upper-tier controllers	off	off	off	**on**

Note what this table settles: the adaptive scaffold runs at *every* tier. Only three blocks are gated, and the two that carry DT-GSK's distinctive claims — the interaction memory and the deterministic polish — do not exist below $D = 50$.

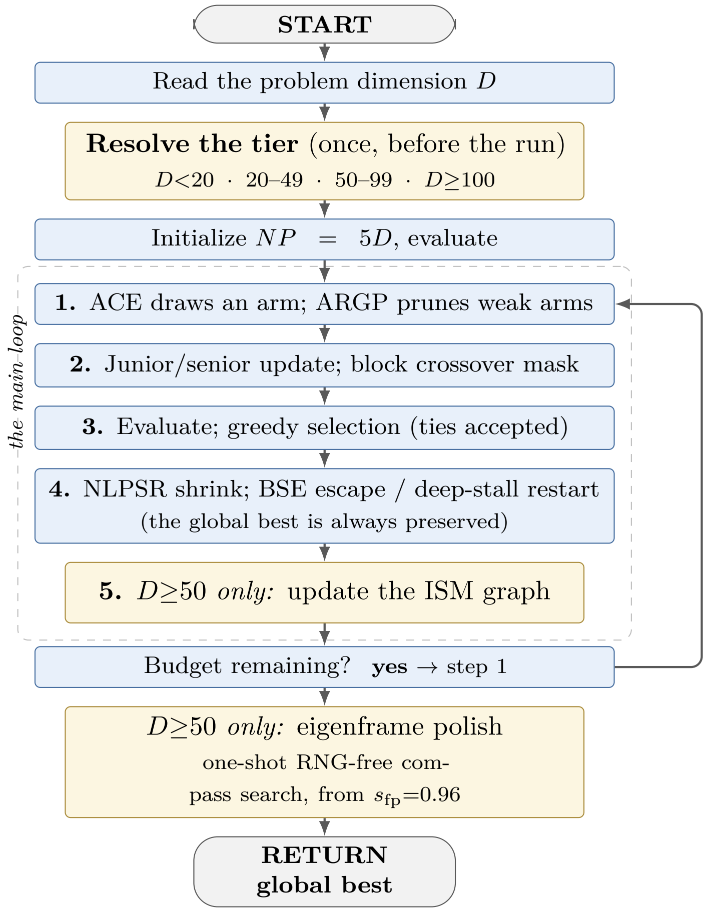

Figure 1. DT-GSK's execution order. Gold marks the dimension-gated stages; the tier itself is fixed before the loop begins.

3. The three contributions

C2 — the dimension-tiered adaptive scaffold

The scaffold is an ensemble of control mechanisms, all active at every tier:

- **ACE**, a bandit-style operator-configuration selector over five arms (six in the mid-tiers, where a differential-evolution arm is enabled). Arm 3 is the GSK-pure setting $(0.5, 0.9, 10)$, so the base method is always reachable. Credit is an exponential moving average of realized improvement, with a probability floor so no arm dies outright.

- **ARGP**, which soft-freezes arms whose windowed acceptance falls below a tier threshold after warm-up — pruning to $\pi_{\min} = 0.05$ rather than removing, so a pruned arm can recover.
- **NLPSR**, a nonlinear population-size reduction schedule with a tier floor ($N_{\min} = 12$, raised to 25 at $D \geq 50$).
- **Linkage block crossover**, replacing the per-coordinate KR mask with a block mask over non-contiguous coordinate subsets for about 70% of the population. Below $D = 50$ the blocks are random; at $D \geq 50$ they are supplied by the interaction memory.
- **BSE**, a triple-trigger stagnation escape (acceptance, diversity and signal floors) firing a Cauchy rescue and, if the rescue fails, an archive-seeded partial restart — hard-capped so it cannot overshoot MaxFES.
- **A deep-stall restart** that fully re-initializes the working population while preserving the global best, so **a restart can never lose ground.**

At $D \geq 100$ a further block of upper-tier controllers activates: basin memory, a subspace floor, late-accept clipping, frozen-streak broadening, and linkage random-mixing.

C1 — the deterministic eigenframe polish

Where the family ends stochastically, DT-GSK ends with a **one-shot, RNG-free compass search** on a learned eigenbasis, opened at $s_{\text{fp}} = 0.96$ of the budget: step along each basis direction, evaluate, contract, repeat. It is budget-exact — every evaluation it spends is charged through the same accounting path — and it **only exists at** $D \geq 50$, where the graph that supplies its basis exists.

The practical consequence is repeatability. A stochastic endgame returns a slightly different incumbent each run; this one returns bit-identical results — *in the declared single-threaded environment the study used.* The paper is explicit that this is not a cross-thread or cross-platform guarantee.

C3's exploratory component — ISM, and its null result

The interaction-structure memory accumulates a decaying ($\lambda = 0.95$), confidence-gated ($\kappa_{\min} = 0.35$) coordinate-pair graph from strictly improving accepted moves. It has exactly two consumers: linkage blocks for the crossover mask, and the signed-graph eigenbasis for the polish. **Like the polish, it is gated to** $D \geq 50$**.**

It costs **no extra objective evaluations** — it is assembled from moves the optimizer already made. It is not free in wall-clock, though: isolating it measured **+57.3% at CEC2017** D=50**, +36.3% at** D=100**, and +30.3% on CEC2013** D=50.

And the direct isolation is a **registered negative result: no detectable standalone benefit at its active tiers.** The authors ran the contrast to find out, got a null, and advertised it in the abstract and conclusions rather than confining it to a supplement. So the component cost real time and bought nothing measurable — and that is the published finding. It is positioned as a secondary exploratory mechanism, explicitly not the performance driver.

4. The method in pseudocode

INPUT: objective f, dimension D, bounds $[\ell, u]$, budget MaxFES (suite-dependent), seed.

1. **Resolve the tier from** D — once, before the run:
   $D < 20$ $\rightarrow$ scaffold only (ACE+ARGP, NLPSR, blocks, LS, BSE, restart)
   $20-49$ $\rightarrow$ + differential-evolution arm in the ACE pool
   $50-99$ $\rightarrow$ + ISM graph, ISM-fed linkage blocks, eigenframe polish
   $D \geq 100$ $\rightarrow$ + upper-tier controllers (basin memory, subspace floor, ...)

2. **Initialize** $NP \leftarrow 5D$ from the seeded stream; evaluate; open the 13 RNG substreams.

3. **REPEAT** while budget remains:
   a. ACE selects an arm (KF, KR, K_{exp}) by EMA credit; ARGP soft-freezes under-performing arms to $\pi_{\min}$ after warm-up.
   b. Partition coordinates junior/senior by the K_{exp} schedule; build the trial vector (junior: near neighbours; senior: best/middle/worst).
   c. Apply the crossover mask — block mask for ≈70% of rows (ISM-supplied blocks if $D \geq 50$, random otherwise).
   d. Repair to bounds; evaluate; greedy selection, accepting ties.
   e. NLPSR shrinks NP toward its tier floor.
   f. *If $D \geq 50$:* update the ISM graphs from strictly improving accepted moves; re-extract linkage blocks on the tier cadence.
   g. *If stagnating:* BSE Cauchy rescue on the worst rows; if it fails, archive-seeded partial restart. On a deep stall, full re-initialization — the global best is never discarded.

4. **Eigenframe polish** *(only if $D \geq 50$)*, once, from $s_{\text{fp}} = 0.96$ of the budget: compass search along the signed-graph eigenbasis — step, evaluate, contract — with no RNG draws and budget-exact accounting.

5. **RETURN** the global best ever accepted.

Steps 1 and 4 are the novelty. Everything else is the family's existing machinery, re-scheduled: the contribution is *where* each mechanism is allowed to act, not a new base operator.

5. How it was tested

A claim like "this method is better" is only as strong as the comparison behind it. This study was built so the comparison can be checked.

Who competed

Seven methods: DT-GSK plus **six published methods from the same family** (GSK, AGSK, APGSK, FDB-AGSK, ATMALS-GSK, and eGSK).

The six were not compared using numbers copied out of their original papers. They were re-built and re-run inside this project, so every method faced exactly the same problems and the same budget.

One disclosure the paper makes, and this summary repeats: five of the six competing methods were written or co-written by two of this paper's own authors, including the original GSK. No method from outside this family is included in any comparison.

On what

Five standard test collections that this research community uses. That came to 12,265 runs per method, or about **86,000 runs in total**.

Collection	What is in it	Sizes	Runs each
CEC2017	29 mathematical test problems	10–100 unknowns	51
CEC2011	22 real-world problem descriptions	as given	25
CEC2013	28 mathematical test problems	10–50 unknowns	51
CEC2020	10 problems, all small	5–20 unknowns	30
CEC2013LSGO	15 very large problems	1,000 (two at 905)	25

Under what rules

- **Equal budget.** Every method got exactly the same number of tests on every problem. Nobody got extra tries.
- **Matched starting points.** On each problem, every method used the same recorded random seed. The six older methods also started from an identical set of candidate answers. **DT-GSK is the one exception the paper documents**: it builds its own starting set from the same seed, because it uses a different group size. The authors flag this as the one place the comparison is not perfectly matched.
- **An analysis plan written in advance — for one collection.** For CEC2020, the authors wrote down exactly how they would analyse the results *before* running them, and saved that plan with a timestamp. That makes it impossible to pick a flattering analysis afterwards. But this covers CEC2020 only. The other four collections had no plan committed in advance, and on the very large problems the authors say they had already seen the standings before writing their plan.

The statistical protocol

Standings are **Friedman mean ranks** over the seven-member panel, taken per dimension and aggregated. Separation is tested with **Wilcoxon signed-rank** tests at the function level, **Holm-corrected** within each dimension family, with Nemenyi critical differences reported for the omnibus layer and rank-biserial effect sizes alongside.

Two disciplines matter for reading the results:

- **The rank aggregates carry no attached test.** The headline cross-dimension figures are descriptive summaries; no cross-dimension test is claimed for them.
- **Evidential tiers never cross.** An omnibus or Nemenyi separation is never reported as a paired-test result, and the CEC2013LSGO descriptive layer — inspected before the analysis plan was signed — carries no confirmatory weight anywhere.

6. What they found

Collection	What kind of test it is	DT-GSK's place	Avg. rank
CEC2017	the collection its settings were chosen on, so the least independent of the five	**1st of 7**	2.48
CEC2013	a second check, but not an independent benchmark	**1st of 7**	2.80
CEC2013LSGO	added later, exploratory; standings were looked at before the plan was written	tied 1st with AGSK	3.13
CEC2011	real-world problem descriptions	2nd, behind eGSK	3.36
CEC2020	the one test planned in advance	**4th of 7** (AGSK 1st, 2.09)	4.11

In short: first on two collections, tied first on one, second on one, and fourth on one.

Reading the good results fairly

Three things limit them.

Who it was compared against. DT-GSK was tested against its own family, not against the strongest optimization methods in the world. The paper makes no claim about the wider field. On the 1,000-unknown problems it says so plainly: specialist large-scale methods report much better published results than any method in this family, DT-GSK included.

No test is attached. The first places are average-rank summaries. Against its closest rival, eGSK, DT-GSK's lead on the main collection is small enough that the statistical tests cannot call it real at any problem size.

Where the settings came from. CEC2017 gave the headline first place, and CEC2017 is also the collection the authors used to choose DT-GSK's settings. Six candidate setups were compared on it before one was frozen. So that result should be read as performance on home ground, not as a fresh test.

The tied first place on the very large problems is the weakest of the three. On that collection the tests cannot tell six of the seven methods apart at all.

Reading the bad results fairly

Second place on the real-world set. On CEC2011, eGSK beat DT-GSK, and the statistical test says that gap is real. One clarification about what CEC2011 is: these are engineering problems taken from practice and turned into a standard benchmark. That shows the method has been tried on a broad range of problem types. It is not evidence of how it would perform in an actual deployment, and the paper is careful about that distinction.

Fourth place on the small problems. On CEC2020, where every problem has 20 or fewer unknowns, DT-GSK finished fourth of seven. The reason is built into the design: its two size-gated parts, the pair-map and the final polish, do not switch on until 50 unknowns. So here it competes without the machinery that wins it results higher up. Its self-tuning machinery is still running, so this is a real result about the method as it is configured, not a fluke.

The authors had written down, before running it, that they expected DT-GSK *not* to lead this collection. That same plan also required them to report the outcome as a real finding rather than explain it away.

Three confirmed losses, not one. CEC2011 is the only confirmed loss for a whole collection. But it is not the only one. On CEC2020 at 20 unknowns, the same test confirms that both AGSK and FDB-AGSK beat DT-GSK. Those two sit in the study's strongest category of evidence, because CEC2020 is the collection that was planned in advance.

The shape of the strength

The advantage is **tiered, and it is not simply "the bigger the better".**

Across the two mathematical collections covering 10 to 100 unknowns, DT-GSK leads at every size except 30, where it drops to second on CEC2017 and third on CEC2013. On the small collection it leads at none of the four sizes. On the real-world set it is second. **Its strength is concentrated at 50 unknowns and above.**

7. Limits, and why you can check all this

What the paper does not claim

- It does not claim DT-GSK is the best optimization method available. It does not even claim DT-GSK is the best method in its own family overall. The paper reports its standing size by size rather than as a single verdict, and other family members beat it at some sizes and on the real-world set.
- It does not claim the pair-map works. The direct test of it came back negative, and that is reported.
- It does not claim to run faster. One competitor, eGSK, could only be rebuilt using a substitute for one internal solver. That part uses some of the same budget and can change the answer it returns, so the eGSK results here belong to this rebuild rather than to the published original. That cuts both ways: eGSK is the method that beats DT-GSK on the real-world set.

- It does not claim the settings are the best possible. They were frozen and never re-tuned per problem, with one disclosed exception: two values were added for the very-large-problem collection after results on it already existed, so that result is not claimed to be blind to its own outcome.

Why the numbers can be checked

The unusual thing about this project is not the method. It is the paper trail.

- **Every run is kept**, stored in read-only frozen data sets rather than summarised away.
- **Results can be reproduced in the setup the authors describe.** The random number streams are pinned to recorded starting seeds, so re-running DT-GSK on a matching setup gives back the same numbers, not merely similar ones. This promise is made for DT-GSK and this project's own analysis tools.
- **Every number in the paper can be traced** back to the data file it came from, and automatic checks confirm the paper and the data still agree.
- **The negative results are published too**: the component that did not work, the computing time it cost, the loss on the real-world set, the fourth place, and the two confirmed losses at 20 unknowns.

If you have thirty seconds to explain it

> It is a search method for hard problems where you can test an answer but cannot work out the best one directly. Its main idea is that it picks its own machinery based on how big the problem is, like choosing a gear before you set off, instead of using one setting for everything. Above a certain size it also finishes with a careful, repeatable polish rather than a random one. It was tested against six other published methods from *its own family* — not against the strongest methods in the field, a comparison the paper does not make — across five standard test collections with equal budgets. It came first on two, tied first on one, second on one, and fourth on one. It is weakest on small problems, where its size-gated machinery is switched off by design, and it loses on the real-world set. One of its three components was tested on its own, cost 30 to 57 percent more computing time, and showed no measurable benefit — and the authors published that result instead of leaving it out.

Sources. All standings are the descriptive average ranks in the project's frozen analysis files. Protocol figures and every scope statement come from the paper itself. This is a companion document and is not part of the submitted package. For the full method, the statistical tests, and the complete results, see the paper and its Supplementary Material.